Lab 05: Set Up Nexus Repository and Configure User Access

Name: Fatima Kalsoom

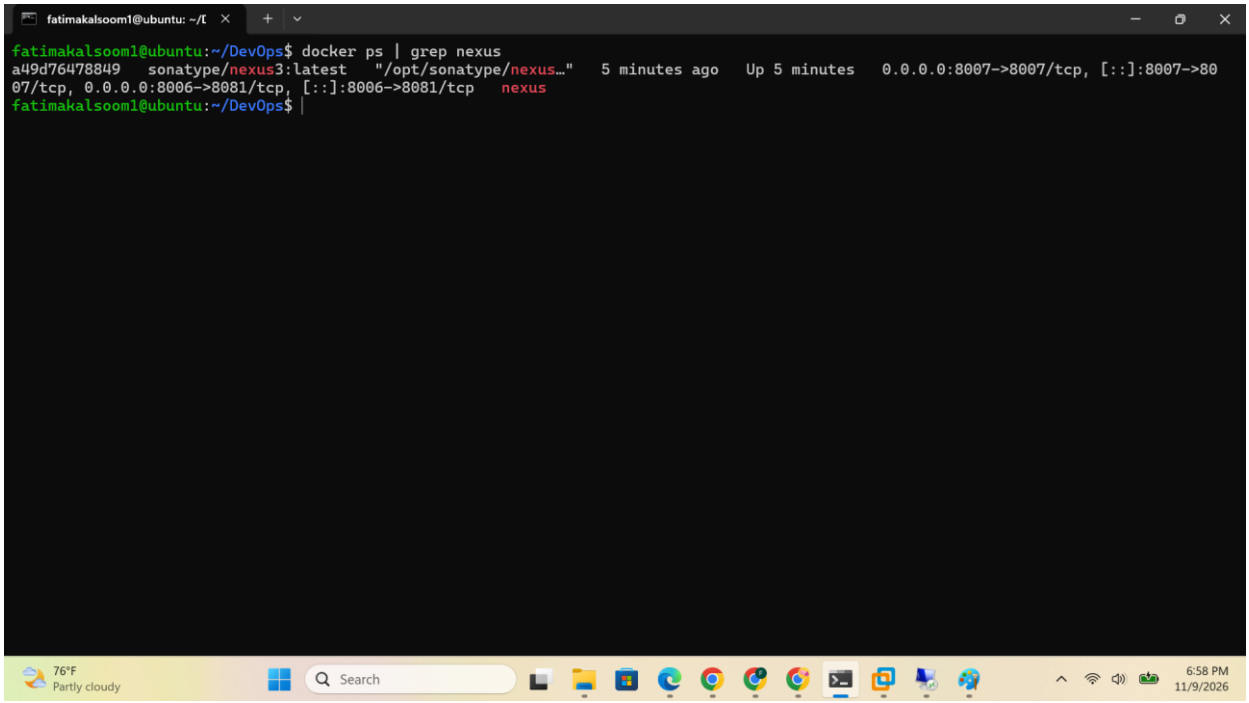
Registration Number: 2023-BSCS-32

Course Code: CS-649

Section: A

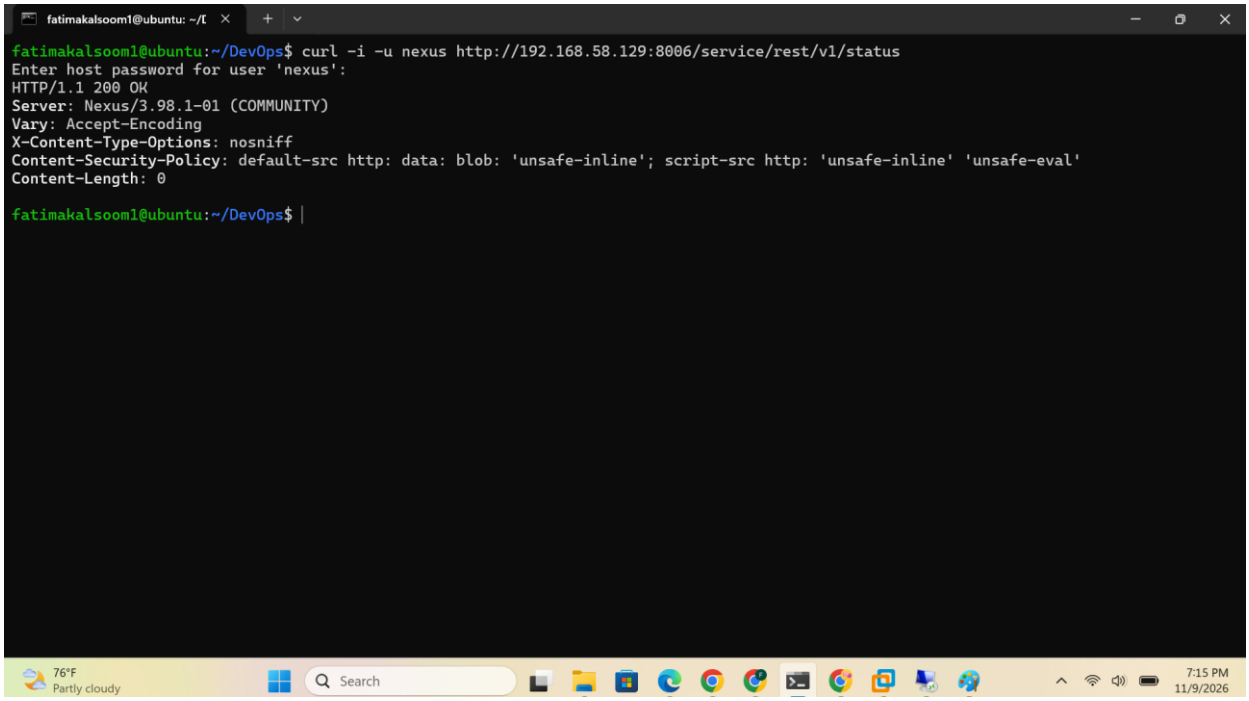
GitHub Username: fatimakalsoom1

Task 1: Run Nexus Repository and Complete the Initial Setup



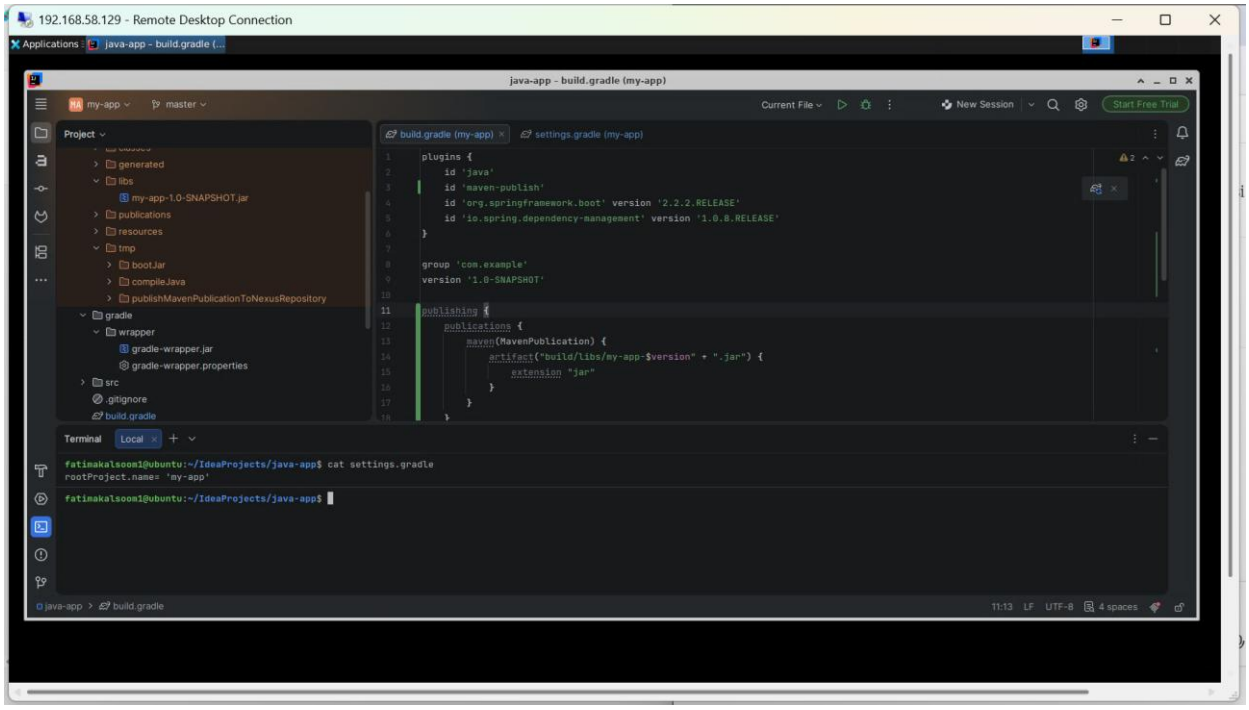
Nexus container running on the Ubuntu server with ports 8006 and 8007 published.

Task 2: Create a Role and User and Verify Access

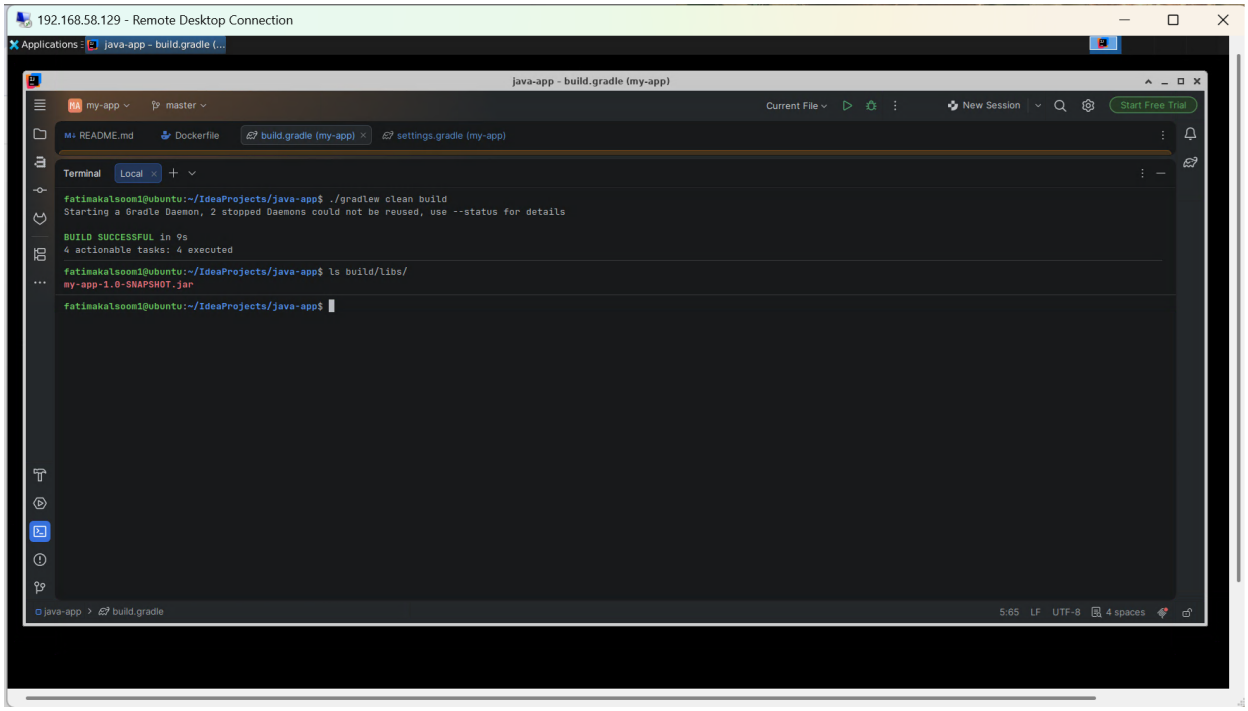


Status API request as the nexus user returns HTTP/1.1 200 OK.

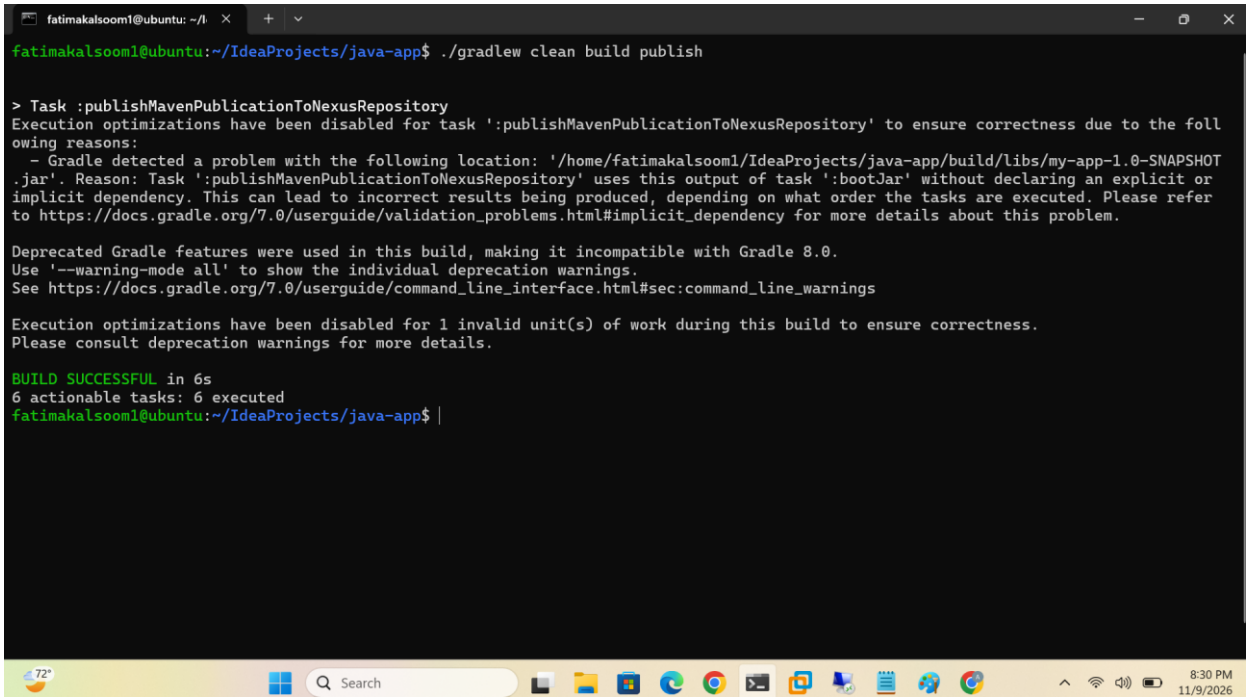
Task 3: Build and Publish the Gradle Application to Nexus



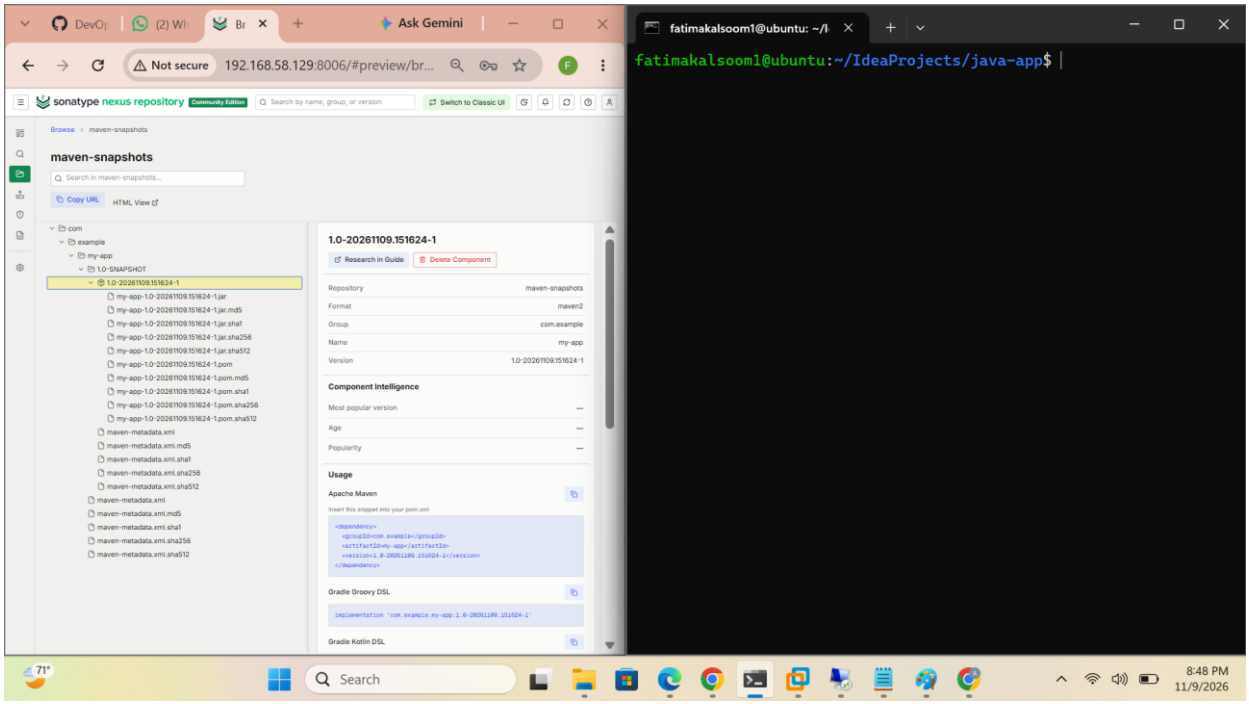
build.gradle publishing configuration and settings.gradle (rootProject.name = 'my-app').



./gradlew clean build succeeds and generates my-app-1.0-SNAPSHOT.jar.

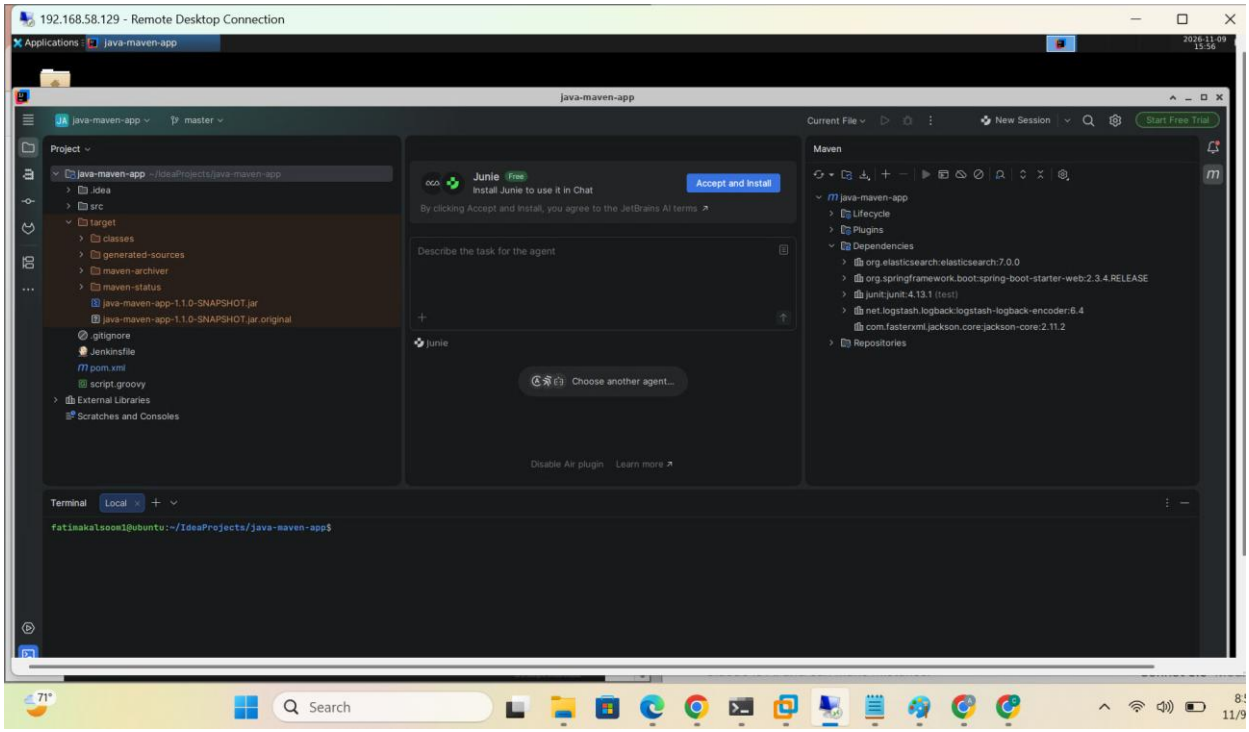


./gradlew clean build publish uploads the JAR to Nexus with BUILD SUCCESSFUL.

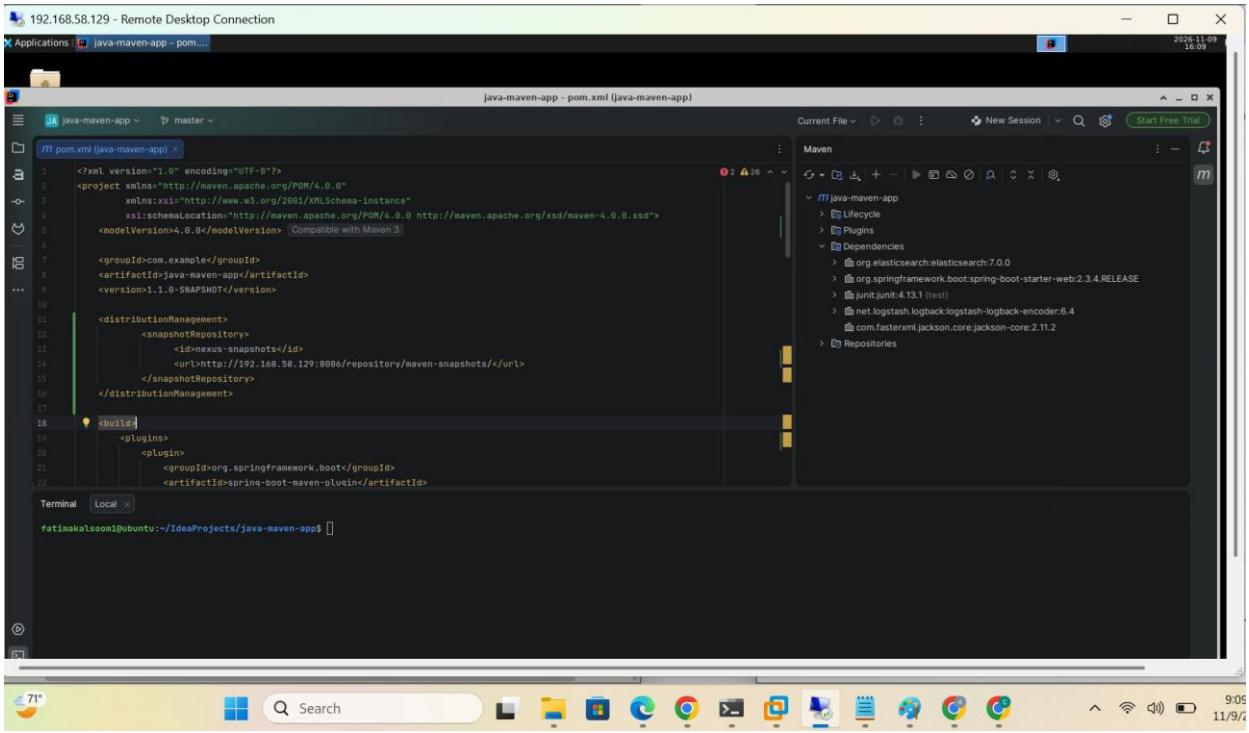


Gradle snapshot JAR and Maven metadata under com/example/my-app/1.0-SNAPSHOT in maven-snapshots.

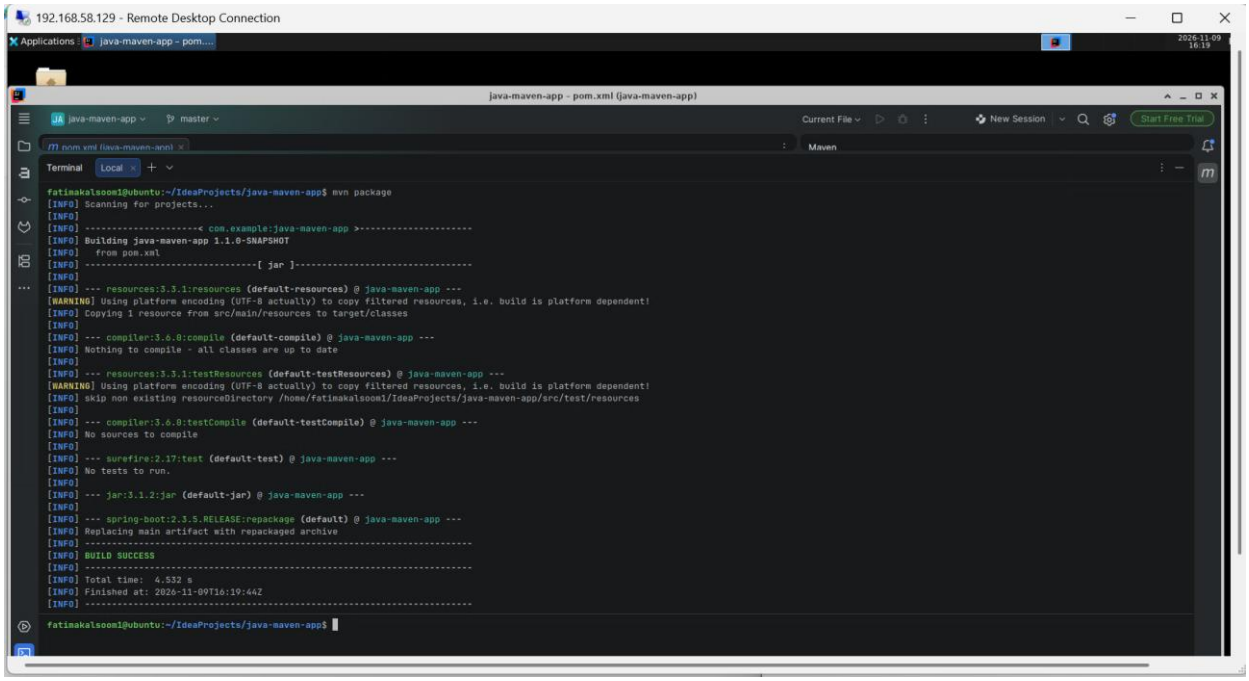
Task 4: Publish the Maven Application to Nexus



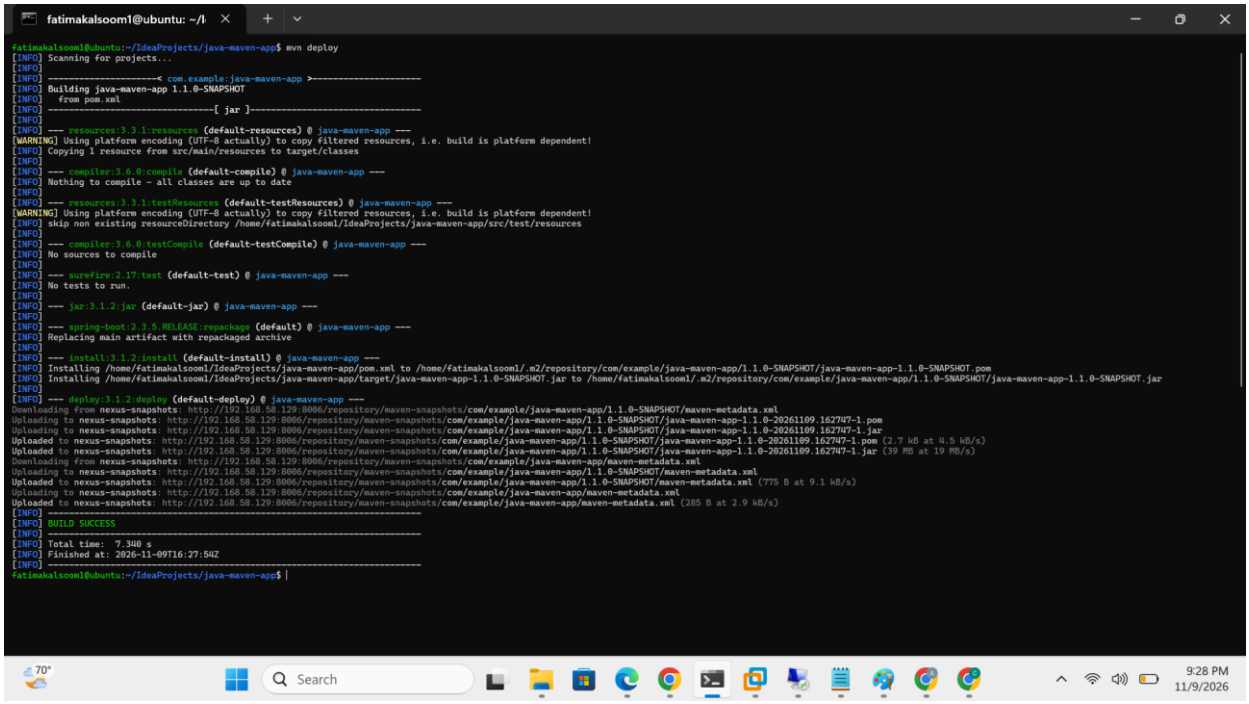
java-maven-app project opened in IntelliJ IDEA.



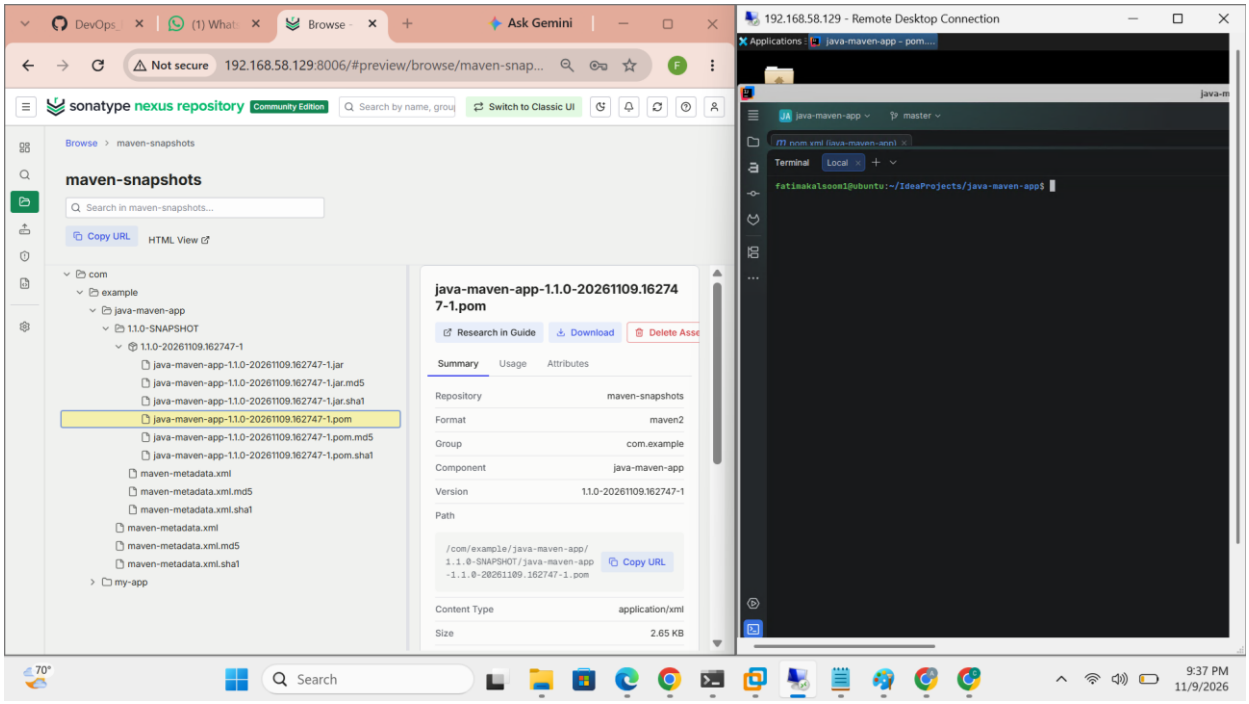
distributionManagement block added to pom.xml.



mvn package completes with BUILD SUCCESS and creates the snapshot JAR.

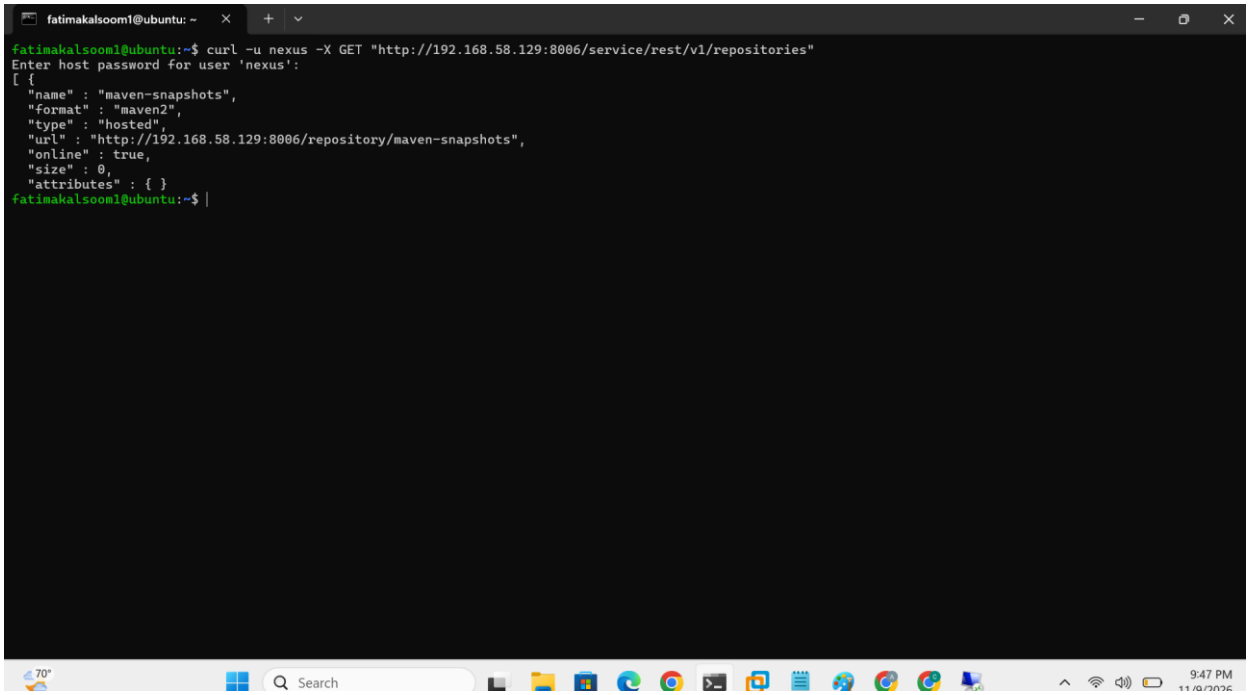


mvn deploy uploads the artifact to Nexus with BUILD SUCCESS.

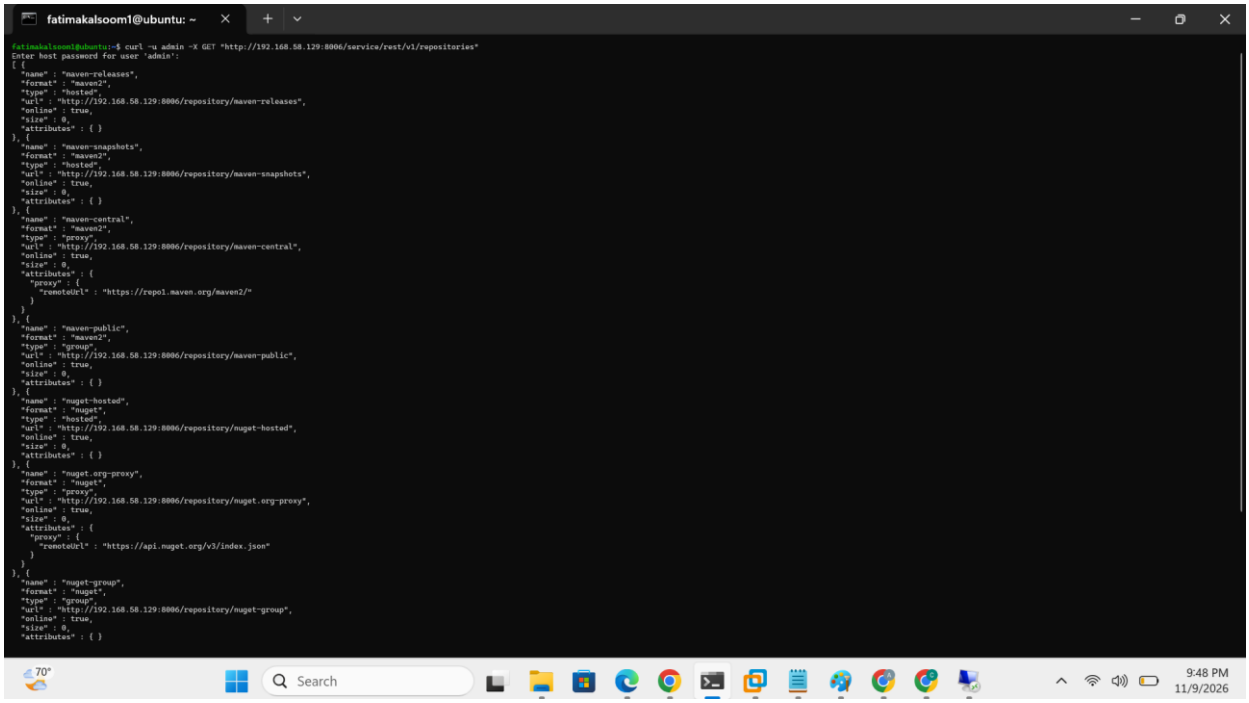


Maven snapshot JAR, POM and metadata under com/example/java-maven-app/1.1.0-SNAPSHOT.

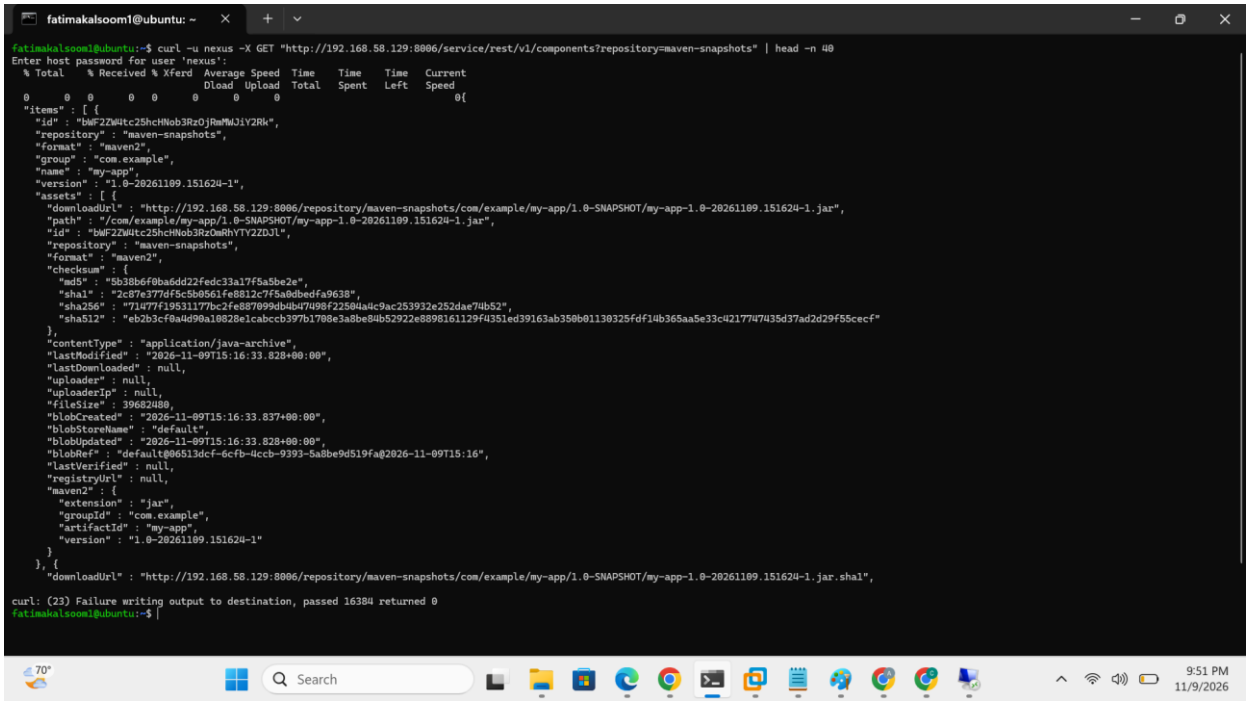
Task 5: Inspect Nexus Repositories and Components Using the REST API



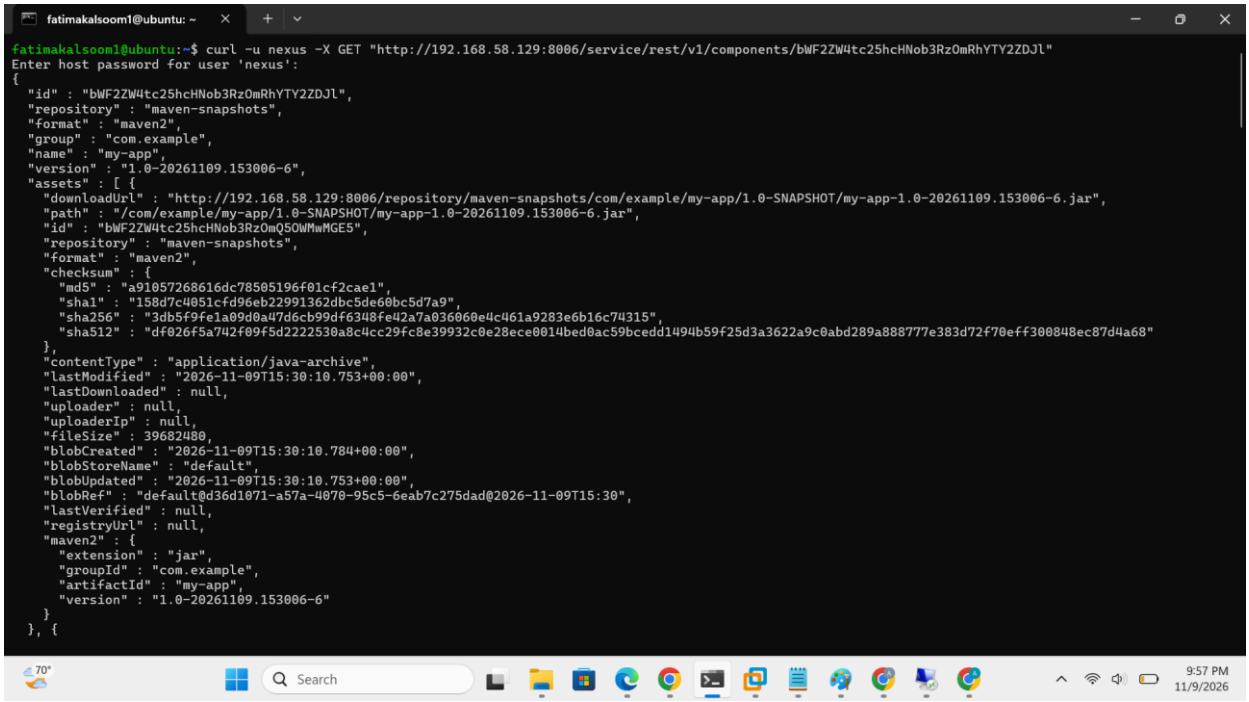
Repository list returned for the restricted nexus user.



Repository list returned for the admin user.

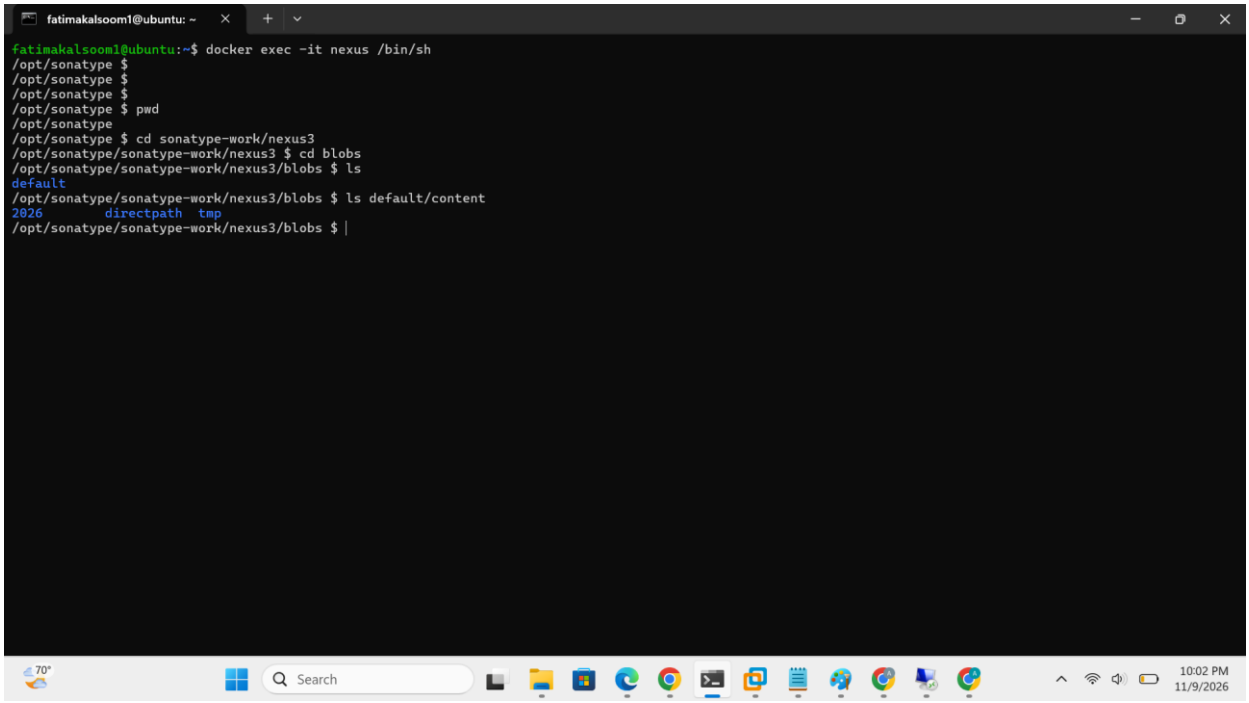


Components in maven-snapshots listed through the REST API.

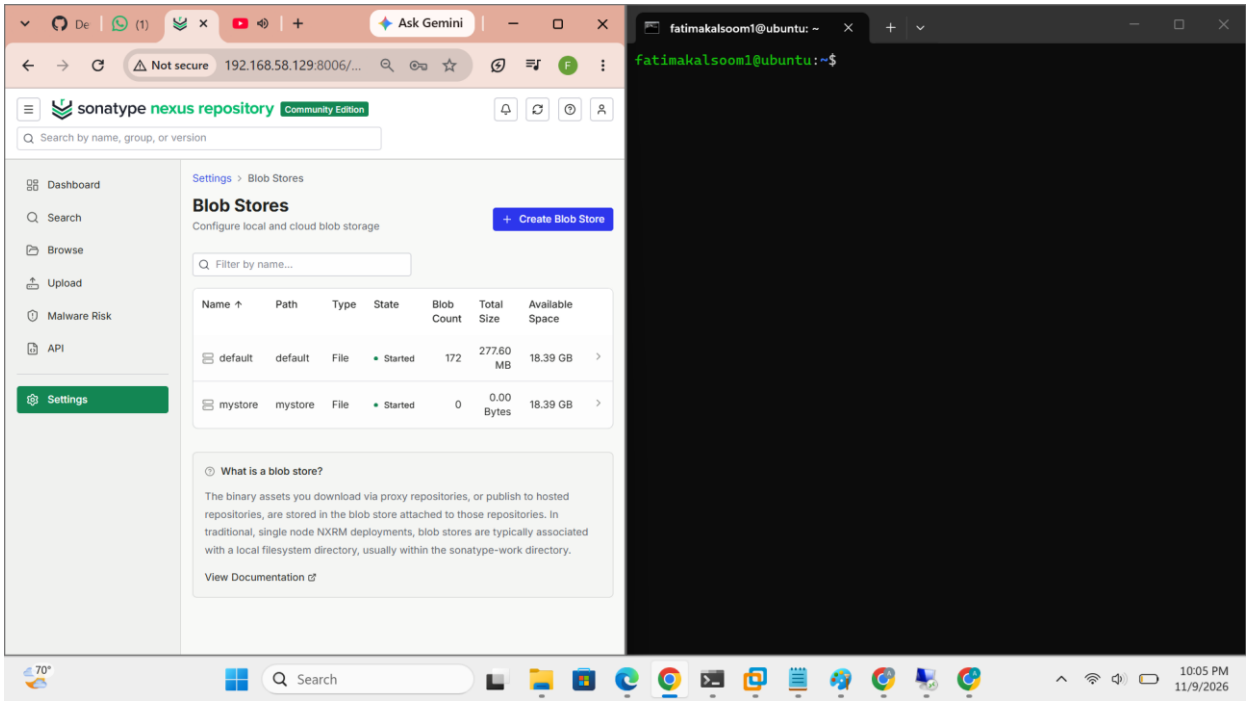


Details of one component retrieved by its ID.

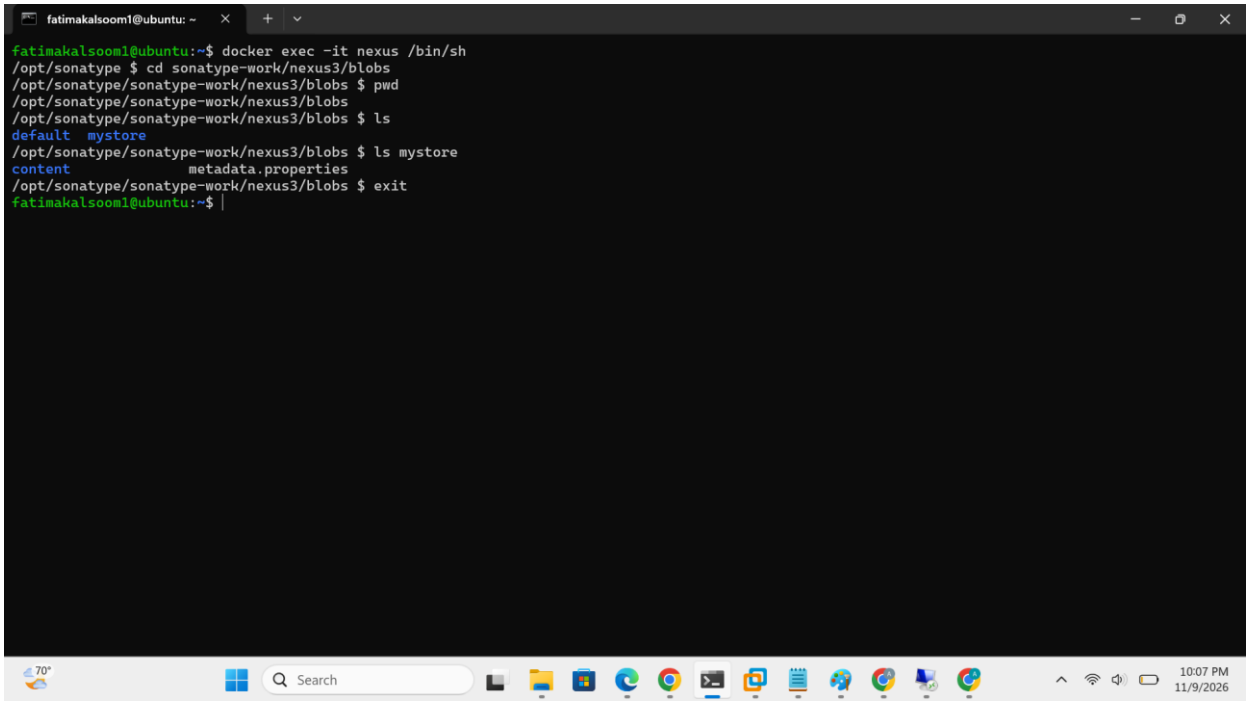
Task 6: Create a Nexus Blob Store and Docker Hosted Repository



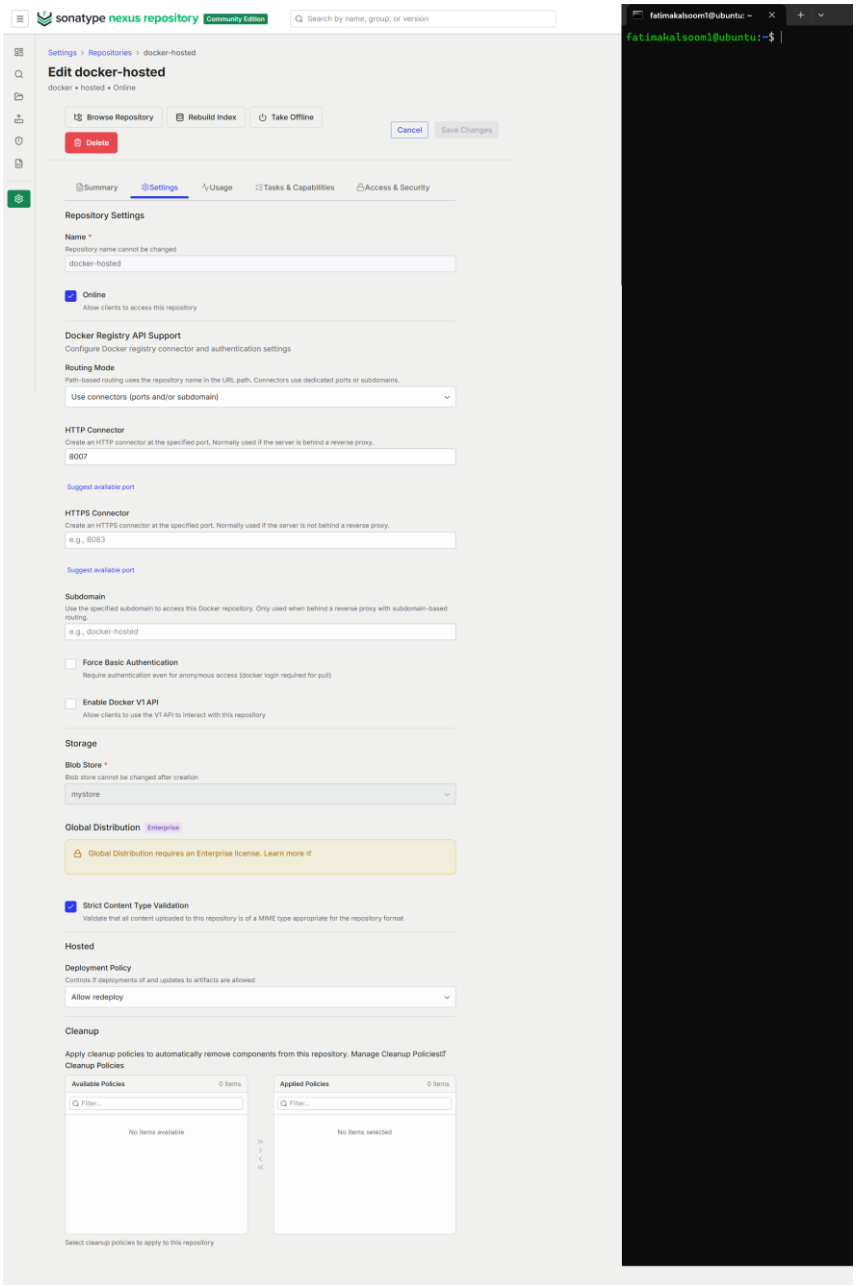
Read-only inspection of the default blob store directory inside the Nexus container.



default and mystore file blob stores both in the Started state.

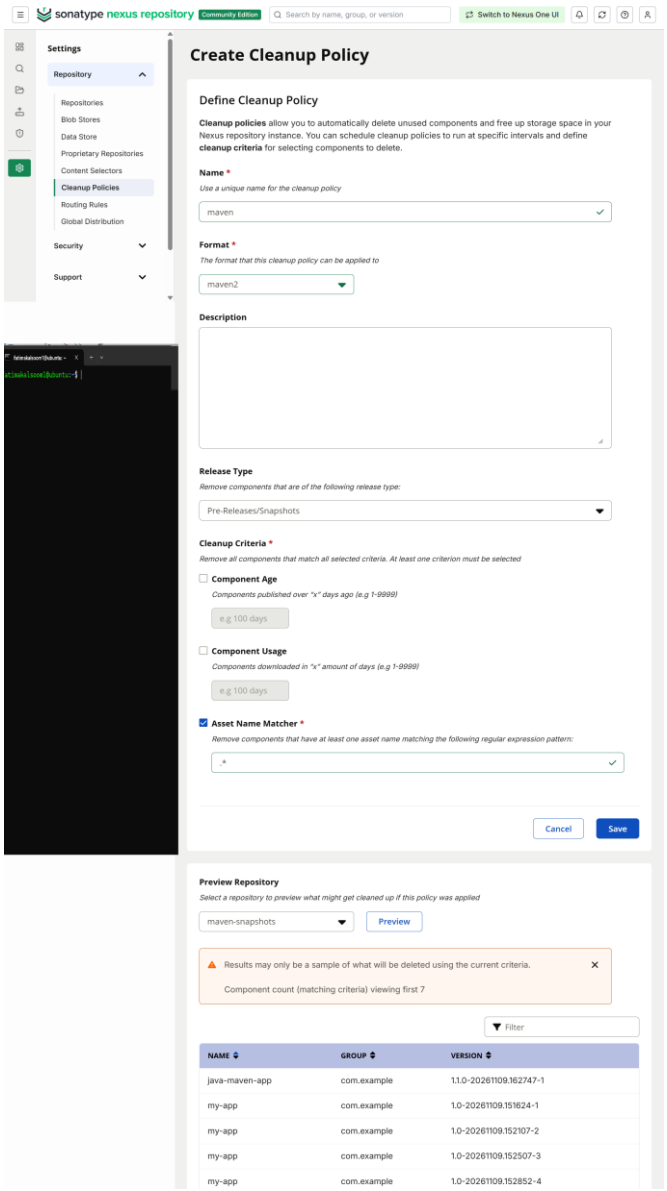


default and mystore directories visible in the container's blobs path.

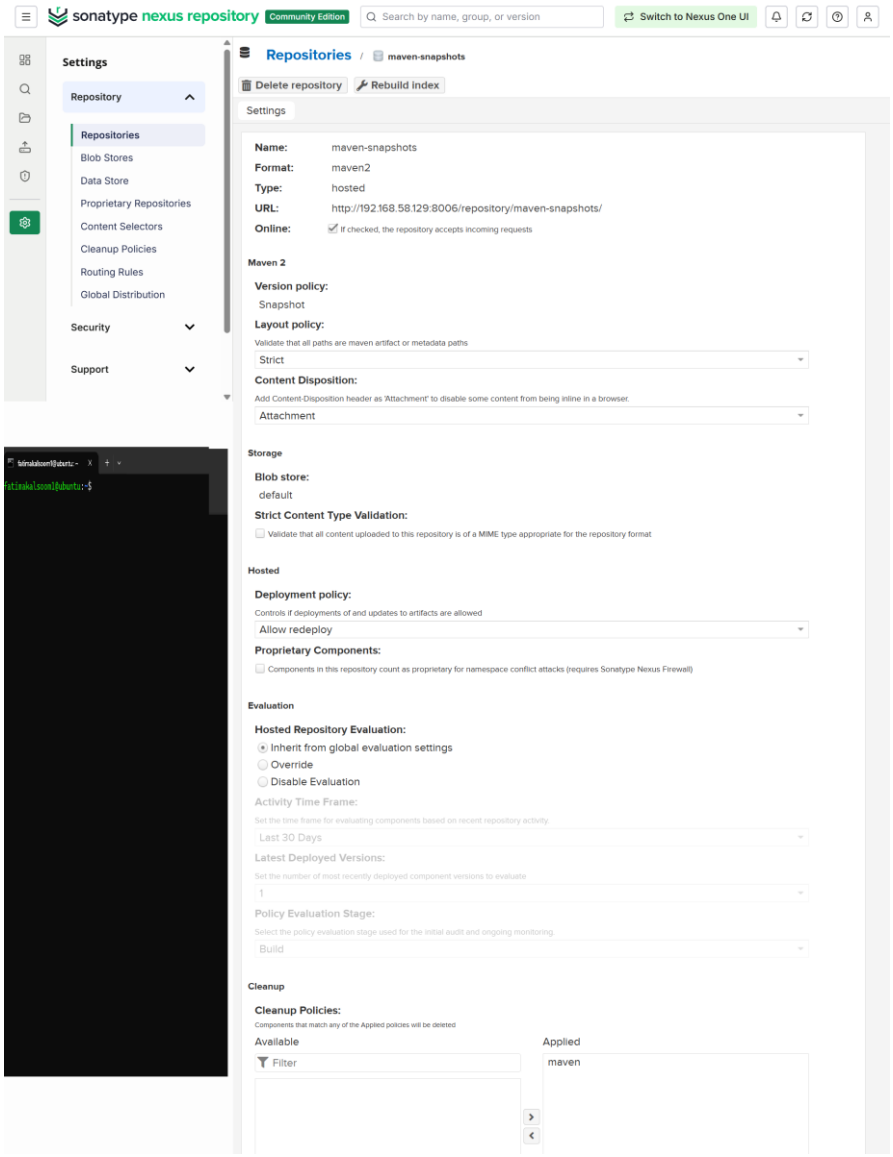


docker-hosted repository using the mystore blob store on HTTP connector 8007.

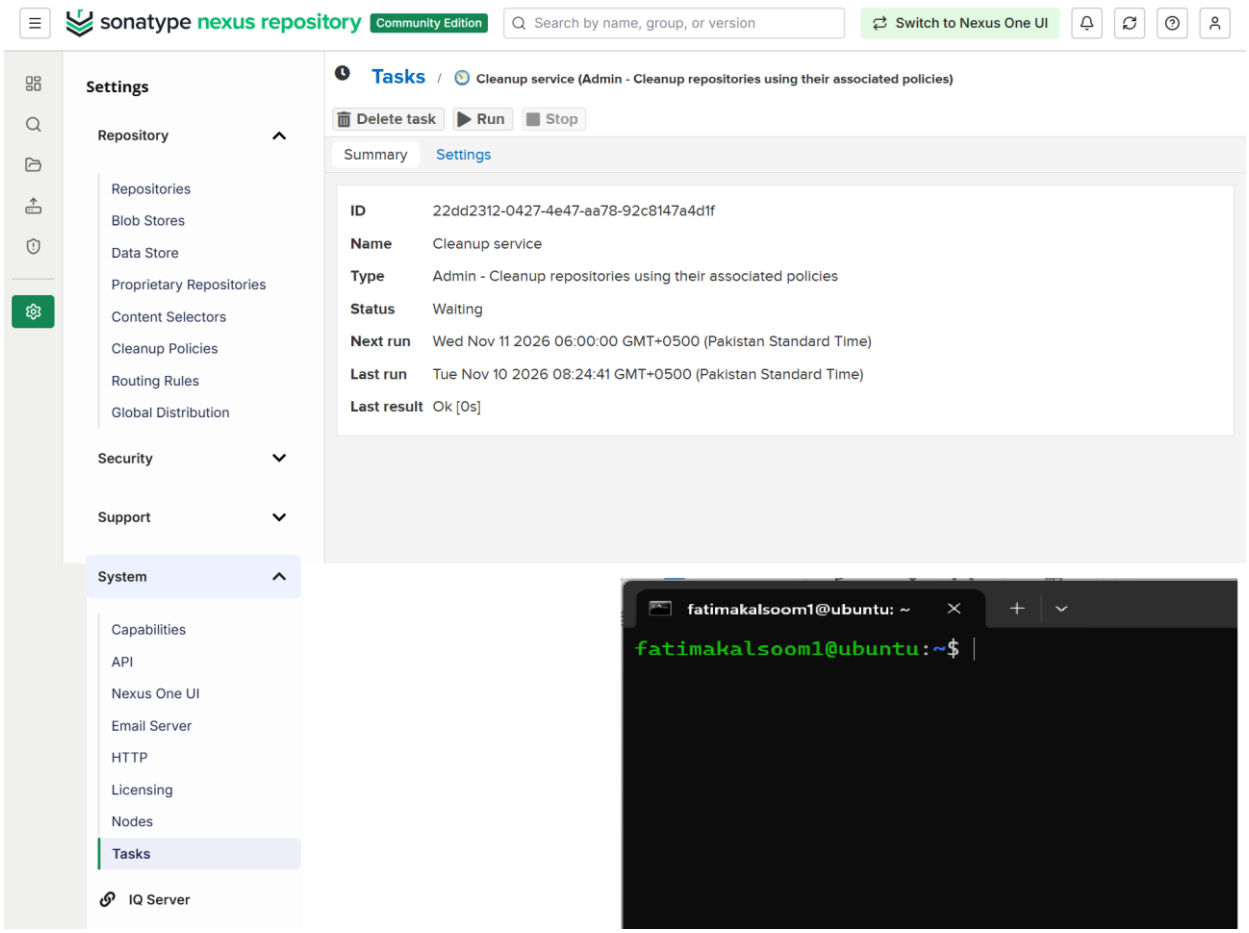
Task 7: Create and Run a Maven Snapshot Cleanup Policy



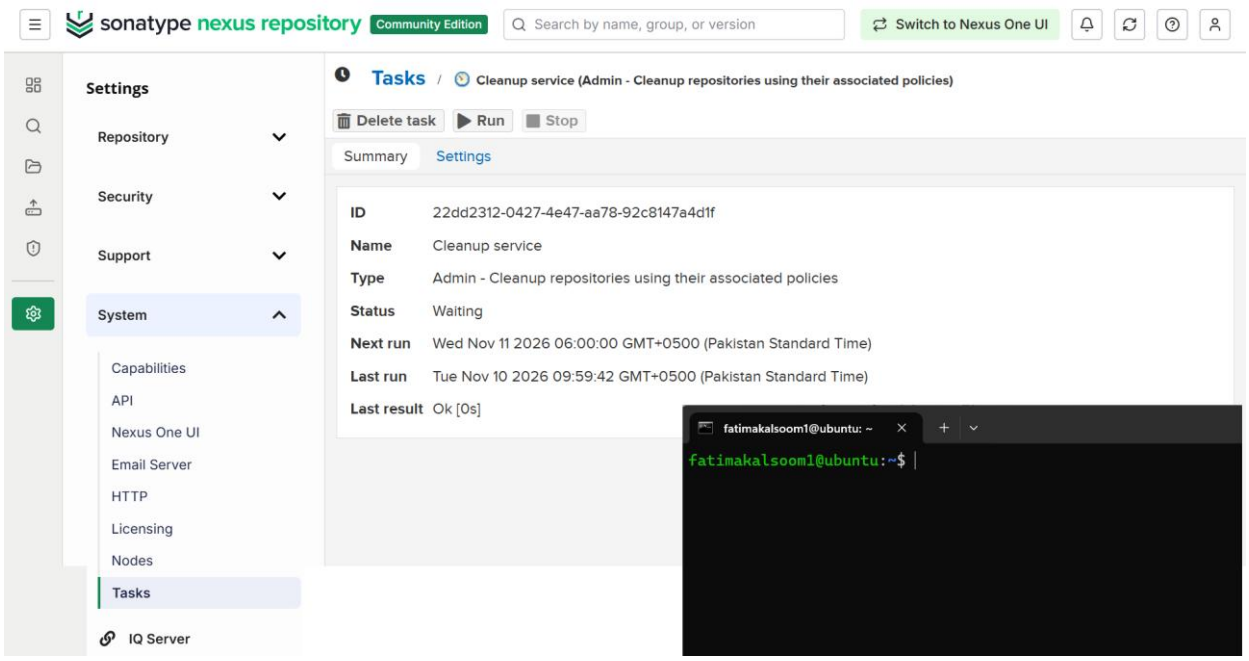
maven cleanup policy (maven2, Pre-Releases/Snapshots, . matcher) with preview of maven-snapshots.*



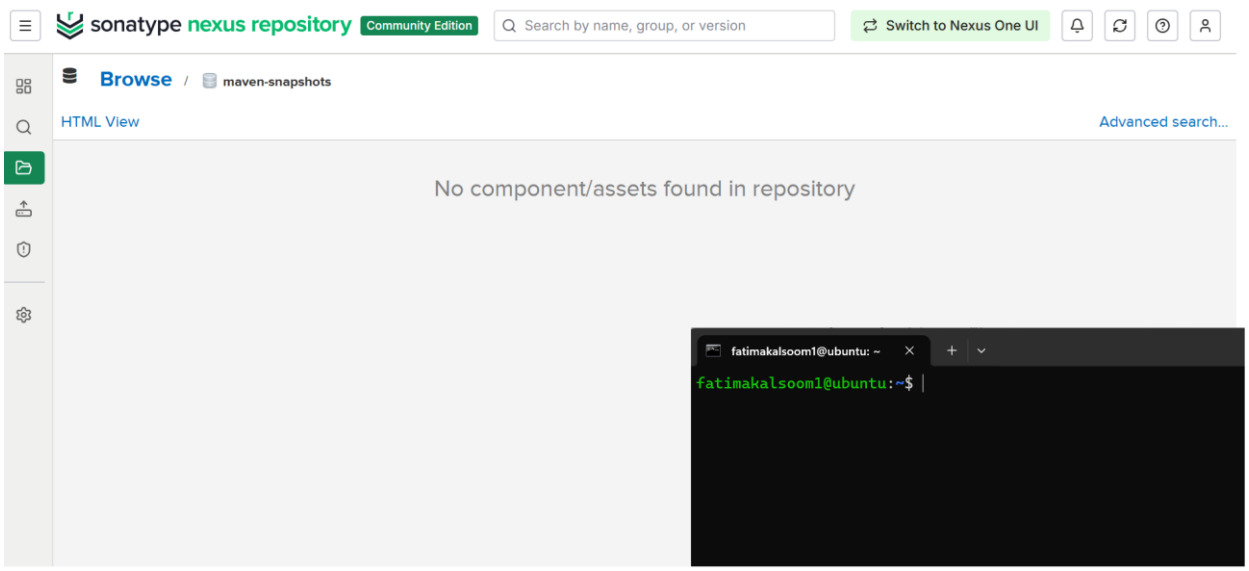
maven policy saved in the Applied list of maven-snapshots.



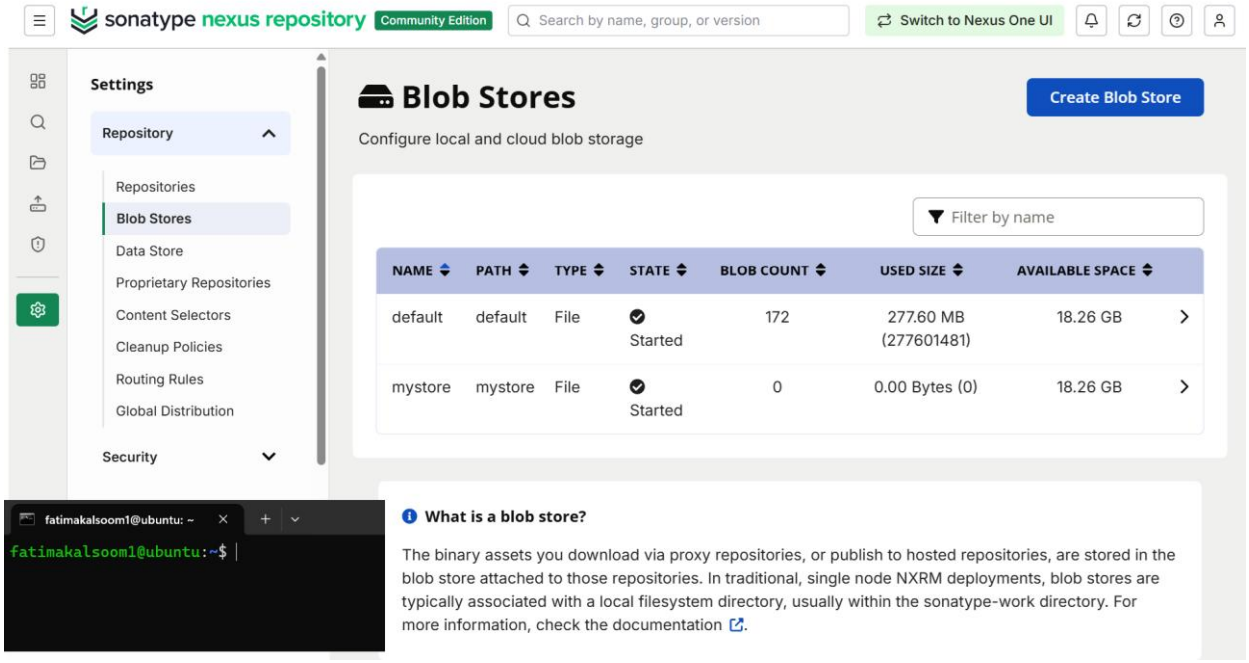
Cleanup service in Waiting state, scheduled for 06:00 the next day.



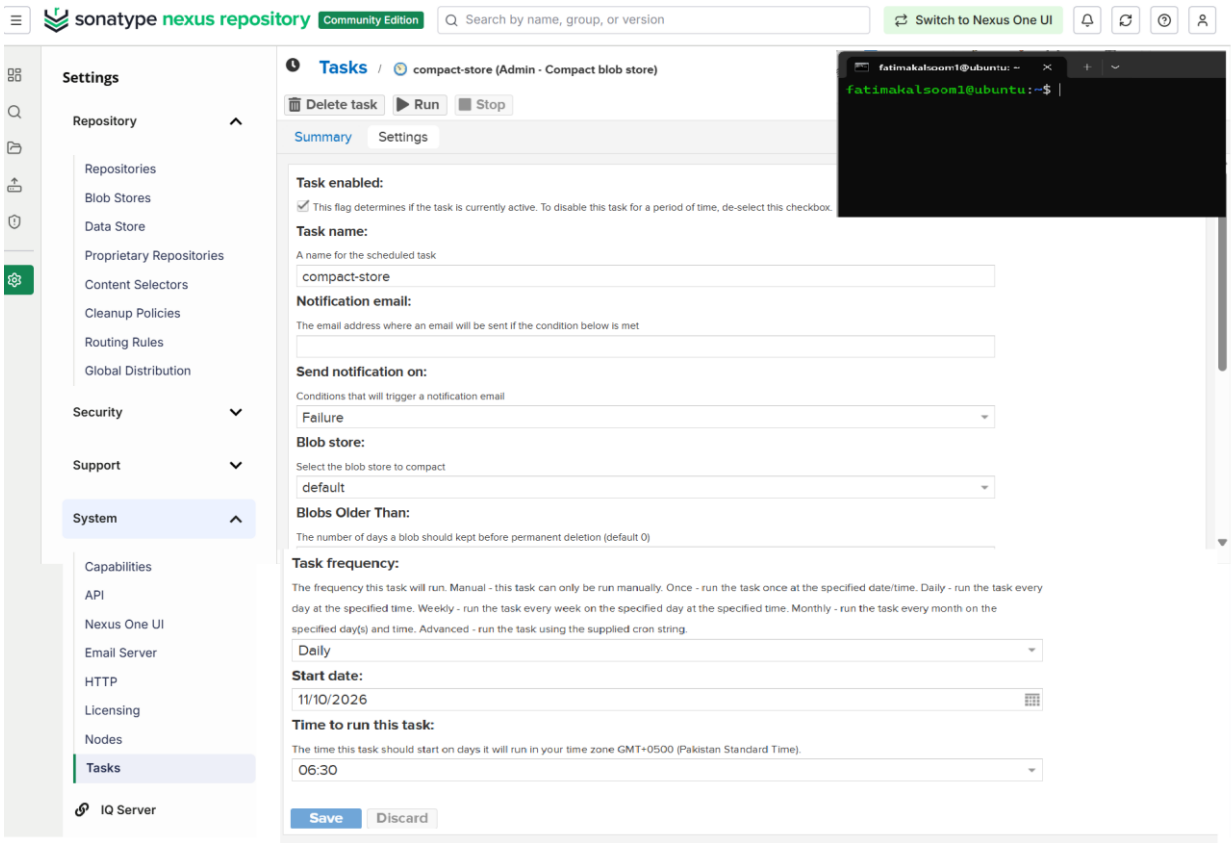
Cleanup service after manual run: Last result Ok with updated Last run.



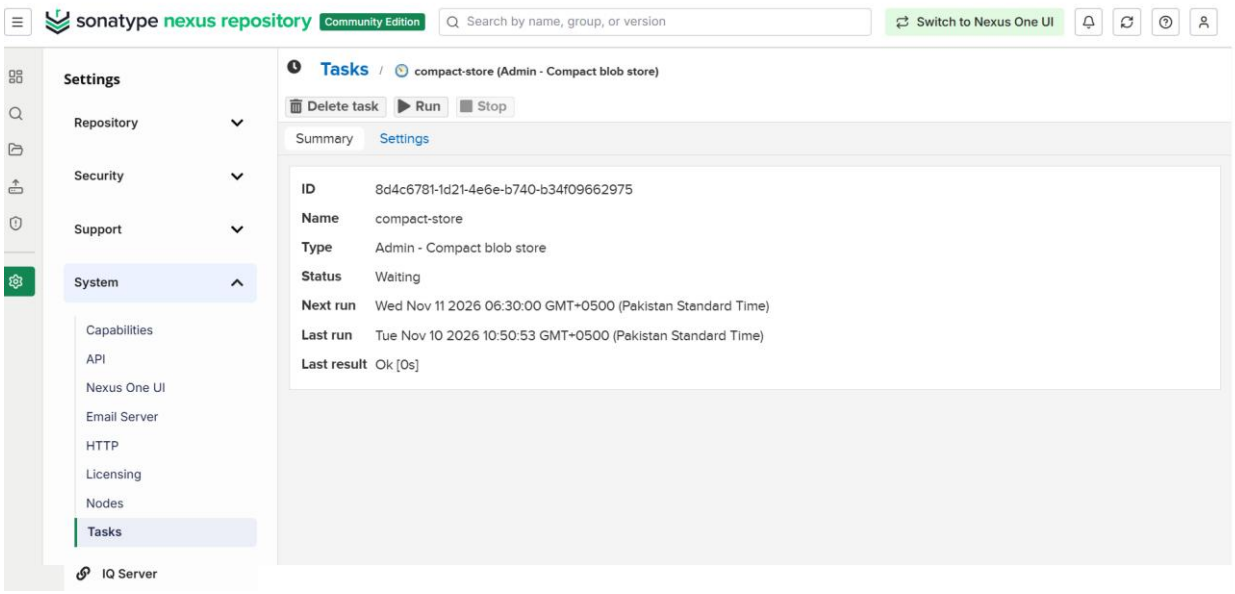
maven-snapshots shows 'No component/assets found in repository'.



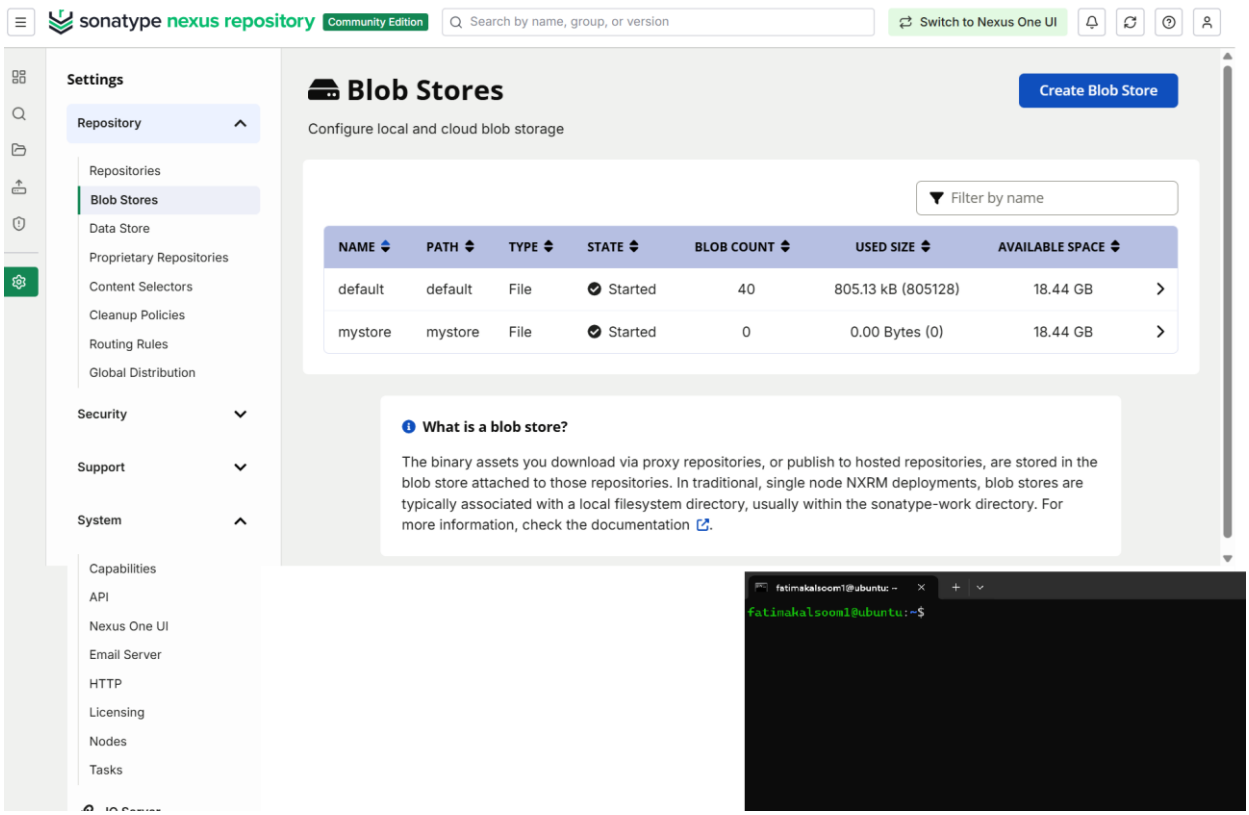
Default blob store before compaction: 172 blobs, 277.60 MB used.



compact-store task: blob store default, 0 days, daily at 06:30.



compact-store after manual run: Last result Ok with updated Last run.



Default blob store after compaction: 40 blobs, 805.13 kB used, more free space.